

CO5-AT1-Q2:

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Course Code: CSA0652

Course Name: DAA for Complexity Analysis

Sudoku Solver (Backtracking / NP-Complete Problem)

Problem Understanding

Sudoku is a constraint satisfaction problem where a 9×9 grid must be completed while following three rules:

- Each row contains numbers 1–9 without repetition.
- Each column contains numbers 1–9 without repetition.
- Each 3×3 subgrid contains numbers 1–9 without repetition.

Algorithm Used

Backtracking with Constraint Checking

Constraints

- A number cannot repeat in the same row.
- A number cannot repeat in the same column.
- A number cannot repeat in the same 3×3 box.

Solution Approach

Steps:

1. Find an empty cell.
2. Try numbers from 1 to 9.
3. Check constraints.
4. Place valid number.
5. Continue recursively.
6. Backtrack if the choice creates conflict.

Constraint Propagation and Pruning

The algorithm reduces search space by rejecting invalid numbers immediately.

Benefits:

- Faster solving.
- Avoids unnecessary combinations.

- Improves efficiency.

Correctness

Every filled cell satisfies Sudoku constraints. The algorithm checks all possibilities and returns only valid solutions.

Complexity Analysis

Worst Case Time Complexity:

$O(9^n)$

Space Complexity:

$O(n)$

Conclusion

Backtracking is an effective method for solving Sudoku and other NP-Complete problems. Constraint checking and pruning significantly improve the solving process by reducing unnecessary searches.